



Strategies for building and managing ‘trust’ to enable knowledge exchange at the interface of environmental science and policy

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ARTICLE INFO

Keywords:

Transparency
Trust repair
Science-policy
Boundary organization
Evidence-informed
Research impact

ABSTRACT

Trust is a critical precondition underpinning successful knowledge exchange among environmental scientists and decision-makers, and thus, evidence-informed decision-making processes. While the importance of trust is well established, however, specific approaches to building, managing and maintaining trust at the interface of environmental science and policy are lacking. Here, we seek to address this gap empirically via in-depth qualitative analysis, using the International Council for the Exploration of the Sea (ICES) as a case study. Specifically, through interviews with members of the ICES secretariat and scientific community, this study aims to (i) explore the importance of trust for organisations working at the interface of environmental science and policy, (ii) identify a range of strategies that are available for building trust at the interface of science and policy, and (iii) identify specific mechanisms for trust repair in situations where trust has been compromised. Our results show that trust is an essential component of a successful relationship between environmental science and policy, although highly dynamic and fragile. For example, while considerable time and effort is required to build trust, it can be lost in only a matter of days. Further, our results show that for successful knowledge exchange, and the attainment of research impact, trust is needed on three levels: (i) trust in individuals, (ii) trust in the organisation, and (iii) trust in the process by which knowledge is generated and exchanged. We identify 14 strategies to building trust, including the need for transparency, the importance of not advocating for a specific outcome, having a clear process for developing and providing knowledge/advice, having regular contact (preferably face to face) and being able to demonstrate independence. Finally, we identify five steps to repair trust: (i) Do not become defensive, be honest that a mistake has occurred, (ii) identify, and explain, why the mistake occurred, (iii) implement measures to ensure that the mistake does not happen again, (iv) correct the mistake (e.g. by providing updated knowledge/advice), and (v) give the trust re-building process time, and ensure ongoing face to face contact throughout that time. In presenting these insights from ICES, this study seeks to provide practical and implementable strategies that can be used by environmental scientists and/or research institutions to help them foster trust with decision-makers, enable improved knowledge exchange and build capacity for evidence-informed decision-making processes.

1. Introduction

Improving knowledge exchange among scientists and decision-makers to enable evidence-informed decision-making processes is a critical component of navigating contemporary environmental challenges that

threaten human well-being and prosperity (Fazey et al., 2013; Cvitanovic et al., 2015a; Hering, 2015). While progress has been made in this regard, for example via the occurrence of ‘bright spots’ whereby environmental science has successfully influenced policy and practice (Cvitanovic and Hobday, 2018), it has mainly occurred at small spatial scales, within

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well-defined contexts and over protracted timeframes. Thus, given the complex, interacting and accelerating challenges associated with pressures such as climate change and biodiversity loss (among others) (Nash et al., 2017; Steffen et al., 2018), there remains a need to further transform the interface of science and policy so as to upscale and accelerate the rate by which environmental science can be incorporated into decision-making processes (Pendleton et al., 2020).

To this end, a growing body of scholarship has sought to more comprehensively understand the barriers that hinder successful knowledge exchange at the interface of environmental science and policy, and strategies for overcoming these barriers (e.g. Cook et al., 2013; Cvitanovic et al., 2015b; Rose et al., 2018; Nguyen et al., 2019; Walsh et al., 2019). A recurrent theme throughout this literature has been the need to transition beyond linear modes of science communication and engagement towards more interactive modes of two-way knowledge sharing (for a comprehensive overview of linear versus interactive engagement see Contandriopoulos et al., 2010). A range of concepts and approaches have been proposed for achieving this. They include, among others, the use of science-policy intermediaries such as knowledge brokers or boundary organisations (Reinecke, 2015; Bednarek et al., 2015, 2018; Cvitanovic et al., 2017, 2018a), the use of participatory research approaches such as those embodied by the notion of co-production (Meadow et al., 2015; Djenontin and Meadow, 2018; Norström et al., 2020), and the implementation of novel institutional structures such as those that ‘embed’ scientists within decision-making agencies, or vice versa (Roux et al., 2019). However, the implementation of such approaches alone does not guarantee improved knowledge exchange (Lemos et al., 2018; Cvitanovic et al., 2019). Rather, a key determinant underpinning their success will be the extent to which they can foster and facilitate relationships among scientists and decision-makers that are founded upon trust (e.g. Hahn et al., 2006; Addison et al., 2013; Clarke et al., 2013; Reed et al., 2014; Gilmour et al., 2015; Cvitanovic et al., 2016; Marshall et al., 2017).

There are various definitions of trust in the literature (see Glenn et al., 2012; Stern and Coleman, 2015), however, in its broadest sense, trust is defined as a psychological state in which an entity (i.e. a trustor) accepts some level of vulnerability based on a positive expectation of another entity (i.e. a trustee; following Rousseau et al., 1998). In this regard, vulnerability refers to the willingness of the trustor to accept and take on some level of risk based on the assumption that the trustee will fulfil their obligations both competently and in a ‘morally correct’ way (Barber, 1983; Mayer et al., 1995; Sharp et al., 2013; Lacey et al., 2015). When considering the notion of trust, it is important to note that a trustor could be either an individual or group, whereas a trustee could refer to an individual, and organisation, an object (e.g. a written policy brief) or a process (Coleman and Stern, 2018). However, as reviewed by Hamm (2017), despite a considerable body of research addressing the construct and idea of trust, there remains a great deal of inconsistency in how it is conceptualised, and consequently measured, particularly in relation to environmental and natural resource management (for an in-depth overview also see Stern and Coleman, 2015; Lacey et al., 2018).

While trust is a critical precondition underpinning knowledge exchange and collaboration, drawing on literature from different disciplines and domains it becomes apparent that high levels of trust might also confer other benefits at the interface of environmental science and policy. For example, in the business management literature, trust is often considered as crucial to organisation success (e.g. Kramer, 1999; Meyerson et al., 2006), increased capacity for problem solving (Nielsen and Nielsen, 2009) and a reduction of transaction costs (the costs required to drive a process towards desired outcomes; Chow, 2008). High levels of trust are also linked to innovation (McEvily et al., 2003), higher levels of stakeholder satisfaction (Brensen and Marshall, 2000) and social learning (Cohen and Levinthal, 1990). While the influence of trust on these features of environmental science-policy interactions are largely underexplored, they suggest that high levels of trust can have a profound influence at the interface of environmental science and policy beyond simply improved knowledge exchange.

Despite the importance of trust at the interface of environmental science and policy, there remains a paucity of information regarding specific strategies for building and managing trust (Boschetti et al., 2016), as well as rebuilding it when compromised. Rather, the majority of literature to date is largely theoretical, with empirical investigations of trust within the environmental sector mainly restricted to understanding how trust operates at the intersection of science and society, or between formal governing agencies and society (e.g. Sharp et al., 2013; Gilmour et al., 2015; Turner et al., 2016; Cvitanovic et al., 2018b). Given the context-specific nature of trust (Hardin, 2002), however, there remains a need to develop empirically grounded guidance for understanding how to build and manage trust for improved knowledge exchange within the context of environmental science and policy.

This study seeks to begin to address this knowledge gap via an in-depth investigation of the International Council for the Exploration of the Sea (ICES) - arguably the world’s oldest and most established organisation focused on improving knowledge exchange among marine science and policy to support evidence-informed decision-making processes. Specifically, through our examination of ICES this study aims to (i) examine the importance of trust for organisations working at the interface of environmental science and policy, (ii) identify and articulate a range of strategies that are available for building trust at the interface of environmental science and policy, and (iii) explore the challenges of maintaining trust and how it can be maintained and rebuilt when compromised (i.e. also termed trust repair, as described in Gillespie and Dietz, 2009). In doing so, this study seeks to provide practical and implementable strategies that can be used by environmental scientists and/or research institutions to help them foster trust with decision-makers, enable improved knowledge exchange, and build capacity for evidence-informed decision-making processes.

2. Methods

2.1. Case study

The International Council for the Exploration of the Sea (ICES) was selected as the case study for this research. ICES was formed as an inter-governmental science organisation in 1902 and operates through the ICES Convention (ICES 1964), and the Copenhagen declaration of strategy (ICES 2002). Its mission is to “advance and share scientific understanding of marine ecosystems and the services they provide and to use this knowledge to generate state-of-the-art advice for meeting conservation, management, and sustainability goals” (ICES, 2019a). This is achieved through a network of over 150 expert groups which engage annually between 1500 to 2000 researchers. This network is supported by the ICES secretariat which is comprised of approximately 60 staff who provide secretarial, administrative, scientific, and data handling support to the ICES community. Funding for ICES is provided by three streams: (1) Each member country contributes annually to the organisation through a “cost share system” which covers the science programmes and the running of the secretariat (~50 % of total income); (2) Process costs to cover the delivery and quality assurance of advice are paid by the requesters of advice (~40 % of total income); and (3) a small proportion of ICES is funded through projects, the ICES Journal of Marine Science and symposia (~10 % of total income). The work of experts within the working groups is not paid by ICES.

ICES has formal agreements or memoranda to provide advice on the management of anthropogenic activities, natural resources, and the conservation of the marine environment. These agreements also require ICES to provide advice that is independent of political influence and subject to best international quality procedures for scientific advice. Thus, the role of ICES as a science advisor has been central to the organisation’s identity and has led to ICES being used as a unique case study for research previously (e.g. Nielsen, 2008; Wilson, 2009; Dankel et al., 2015; Ballesteros et al., 2018). Within the ICES structure, the key group responsible for supporting the translation of ICES science into advice in the Advisory Committee (ACOM).

The importance of trust has been acknowledged in the context of ICES and their end-users (i.e. requestors of advice; Wilson, 2009; Glenn et al., 2012). Indeed, there are numerous examples where ICES lost or eroded trust of fishing industry or managers. Examples include advice on fishing for Northeast Arctic cod (1986), blue whiting (2000s) and mackerel (2010s) (Hoydal, 2014; ICES, 2019b). Also occasions where ICES lost trust by overstepping their role as the providers of evidence and undermining their operational legitimacy, e.g. setting up normative objectives for fisheries management in the 1990s and independently applying principles for the precautionary approach in the early 2000s (Hoydal, 2014). However, over time ICES rebuilt its reputation as a trusted partner to providing knowledge into evidence informed decision making for marine management and now has a growing suite of organisations requesting scientific advice. Thus, given ICES history and reputation, it represents an ideal case study to explore the dynamics of trust.

2.2. Participants

Given the large size of the ICES Secretariat and broader ICES community as described above, for the purpose of this study we focused specifically on ICES members who are directly involved in the translation of ICES science into advice. This included (i) employees of the ICES secretariat (i.e. those employed by ICES who are responsible for overseeing ICES processes) and (ii) annually active members of the ICES community (i.e. key members of ICES working groups to develop advice). Interviews were conducted over Zoom and Skype, as the timing of this research coincided with the Covid-19 pandemic which prohibited face-to-face interviews and travel. All interviews were audio recorded and professionally transcribed to ensure their accuracy.

A deliberate snowball sampling technique was employed for this study (Noy, 2008). The initial group of participants were identified by the research team in collaboration with the Chair of the ICES ACOM on the basis of diversity (gender, geographical location, discipline and career stage) and in recognition of their contribution to ICES operations. Sixteen participants were identified through this process, with a total of 13 taking part in the research. This was supplemented by seven participants identified through the snowballing technique. Initial participants were asked to share contacts from the ICES staff or within the ICES community that they believed would be relevant to the study. This step was repeated at the end of each subsequent interview and prospective participants were invited to participate. Sampling continued until emerging concepts had been fully explored and new data no longer provided new insights, dimensions or theoretical categories (i.e. reaching ‘theoretical saturation’; Bryman 2012; Saunders et al., 2018). Theoretical saturation was achieved after 17 interviews, however, in total we conducted 20 interviews to ensure the robustness and adequateness of our data. This is in line with previous research (e.g. Guest et al., 2006; Baker and Edwards, 2012). Interviews were conducted between July and September 2020.

Ethical approval was obtained from the Australian National University Human Ethics Committee (Ref: 2020/244), prior to the commencement of data collection. Before the interview, the purpose of the research was explained to each prospective participant and formal consent to participate was obtained. In accordance with the approval protocol, and to protect the anonymity of study participants, demographic characteristics of the participants cannot be disclosed.

2.3. Research instrument

A qualitative research approach was adopted in this study to provide a more-in depth and comprehensive exploration of the three study objectives. Qualitative approaches have been employed in analogous research investigating the environmental science and policy interface (e.g. Reed et al., 2014; Marshall et al., 2017; Norström et al., 2020; Cvitanovic et al., 2018a, 2018b), including research into trust and transparency (e.g. Glenn et al., 2012; de Vries et al., 2014; Gilmour et al., 2015). Interviews were structured using an interview guide which was designed to directly explore

the perceptions of participants against each of the three study objectives. Questions pertained to: (i) the importance of trust in ICES as a provider of scientific knowledge and (ii) the importance of transparency when providing advice to requesters (full interview guide in Appendix A).

The interview guide was developed by drawing on existing published evaluations of knowledge exchange activities (e.g. Reed et al., 2014; Cvitanovic et al., 2016, 2017 and 2018a) and was adapted to address the aims of this study. Qualitative pre-testing was undertaken to ensure the adequacy of the interview guide (e.g. providing insights into participants’ comprehension of the materials). The guide was piloted among the research team and one member of the ICES secretariat, and refined accordingly. Minor changes were made to the guide to improve the clarity and context specificity of questions. This was particularly important, as for many participants, English was not their native language. Interviews were undertaken by three of the authors (CC, RS and MM).

2.4. Data analysis

All interview transcripts were analysed using NVIVO 11 qualitative data analysis software. While the research objectives formed the basis of the coding, the analysis of the raw data was completed following an inductive approach, based on Grounded Theory Analysis (Glaser and Strauss, 1967; Charmaz, 2008). This inductive approach enables research findings to emerge from the frequent, dominant or significant themes inherent in raw data (Thomas, 2006) and there are two main stages or cycles (following Blythe and Cvitanovic, 2020).

First, the raw data (interview transcripts) were broadly coded against the three research objectives of this study. Each transcript was coded against a set of descriptors designed to identify emergent themes and to capture the key elements of these themes (Bryman 2012). The use of participants’ own words to derive and summarize key themes (“in vivo” coding) enables research findings to emerge from the interviews without the restraints imposed by structured methodologies (Cvitanovic et al., 2018a, 2018b; Charmaz and Belgrave 2012). Hence, the results are a direct reflection of the language and words commonly used by the research participants, opposed to the potentially subjective interpretations of the research team.

Second, thematic coding was used to identify and interpret patterns or ‘themes’ in the qualitative data that were relevant to the research objectives. This developed a coherent synthesis of key themes (Saldana, 2015; King et al., 2018). Across the two cycles of data analysis, the evolving interpretations were continually verified against the raw data from which they were derived to ensure validity and relevance of themes (see Fleming and Vanclay, 2009; Marshall et al. 2011; Norström et al., 2020).

2.5. Methodological limitations

To assist in the interpretation of the results of our study, it is important at this stage to also note an important methodological limitation. Specially, while case-study research approaches are well recognised for their ability to provide in-depth understandings of complex and/or complicated issues (for a detailed description of case-study qualitative research see Starman, 2013), they are also limited in that the findings cannot be directly extrapolated to each and every context. Specifically in relation to this study, we acknowledge that the interface between environmental science and policy is highly context specific across cultures, space, sector and time. Thus, whilst the strategies presented below for building and maintaining trust have proven critical within the context of ICES, their applicability to other contexts may be more limited. Thus, the results of this paper should be seen as an opportunity to more deeply understand the principles and nuances of trust at the interface of environmental science and policy, and not seen as definitive list of strategies that can be applied in in every situation.

3. Results

The coding of the interview data resulted in 22 themes which were mapped to each of the three study objectives. Themes are presented in Table 1 as an ‘analysis hierarchy’ (i.e. ordered from the themes most frequently mentioned to least frequently mentioned) to provide an overview of the coding results. However, it should be noted that frequency simply refers to the number of times each theme was mentioned by participants, not the level of importance that participants attributed to the specific issue. Specific themes, within each of the three research objectives, are detailed in the following sub-sections.

3.1. The importance of trust

All twenty participants of the present study considered trust as critically important for the success of ICES, and in particular, for knowledge exchange and the attainment of research impact (Table 1). This was highlighted by one participant who stated that trust is: ‘...the alpha and the omega. It’s everything. Without trust, you don’t have anything’ (ID4). The strength of this sentiment was reiterated by another participant who said: ‘It is vital, absolutely critical. I just don’t see how the [science policy interface] can function if trust isn’t part of the equation’ (ID 16).

The analysis of the interview transcripts identified three levels at which trust is important. The first was trust between individuals (e.g. an individual employee of ICES and an individual requestor of advice), which was considered critical for the formation of relationships that facilitate knowledge exchange. As highlighted by one participant: ‘If you are providing guidance or advice... You must have their trust, because if you

Table 1
Analysis hierarchy of themes derived from interviews with research participants (n = 20).

Research Objective	Theme	Frequency ^a	Number of sources ^b
Understanding trust	a Importance of trust	21	20
	b Fragility of trust	21	15
	a Be transparent	55	18
	b Do not advocate	17	14
	c Have regular contact	13	7
	d Be able to demonstrate independence	11	9
	e Acknowledge risks	10	8
	f Ensure data quality control	8	7
	g Independent review of advice	7	7
	h Do not defend advice	5	4
	i Give it time	4	4
	j Ensure those generating advice have correct ‘expertise’	4	4
	k Listen and accept feedback	3	3
Strategies for building trust	l Communicate success	3	3
	m Deliver what was requested/required	3	3
	n Be mindful of politics and political sensitivities	3	3
	a Be honest about the mistake	18	12
	b Identify, and explain, how it happened	9	9
	c Implement steps to prevent it happening again	7	6
	d Correct the advice given	6	6
	e Give it time	6	4
	f Ensure face to face contact	2	2
Trust repair			

^a Frequency refers to the number of times a theme was coded across all interview transcripts.

^b The number of sources represents the number of unique interviewees (i.e. participants) who raised the theme during the interview process (maximum potential N = 20).

lose it then you are no longer of any use’ (ID 14). Participants of our study also suggest that, within the context of ICES, high levels of individual trust can facilitate knowledge exchange by providing space for open dialogue: ‘It has to be high at the individual level...because then there is a frankness and openness in the relationship, and people tend not to hold back from raising an issue when it arises.’ (ID 6).

The second, and most commonly discussed level at which trust was discussed by participants, was in relation to the organisation (i.e. trust in ICES as an organisation). For example, one participant noted that: ‘Trust is important. If stakeholders felt that ICES was ‘sold out to’ environmental NGO’s or the fishing industry, the relationship would be diminished very quickly’ (ID 17). In such cases where organisation trust is lost, another participant noted that: ‘...the organisation loses all of its credibility, all of their legitimacy’ (ID18).

Finally, and closely related to organisational trust, the participants of this study emphasised that there must also be trust in the process of generating and exchanging knowledge. This was emphasized by one participant who said that: ‘It is key...that we have trust in the process...they need to trust that we will deliver the best knowledge that we have at that point in time based on the information that is available’ (ID 5). Participants also noted the steps that ICES had taken to ensure trust in their process (for example, by ensuring as much transparency in the process as possible), as discussed in section 3.2 below.

While discussing the importance of trust at the interface of science and policy, 15 participants also noted the highly fragile and dynamic nature of trust (Table 1). For example, one participant shared comments that were made to them by an external stakeholder: ‘ICES might be in a trusted position now, but don’t forget, one huge mistake and you can lose it, or one small mistake that grows into a bigger mistake, and you will lose it’ (ID 2). While participants noted the considerable time needed to build trust at the interface of science and policy, they also noted the much quicker speed in which it can be lost or eroded: ‘It is damaging, when we give advice and later find out it was wrong...even though we quickly issue a correction, within two days this has undermined their trust in us, and our credibility’ (ID 5). In such cases where trust is lost, participants also noted flow on impacts to organisation legitimacy and credibility, and the ongoing nature of the relationship: ‘If they don’t trust the advice they are given, we would become irrelevant...they would turn to other types of knowledge providers to get their advice’ (ID 9). Further, participants noted that despite best efforts to maintain trust, it will inevitably be challenged at some point in any relationship at the interface of environmental science and policy, as highlighted by this participant: ‘...we are all human beings, we all make mistakes at some point’ (ID 15).

3.2. Strategies for building trust

Data analysis identified 14 specific strategies and considerations for building trust (Fig. 1). Of these the most frequently discussed was the need for transparency across all processes related to knowledge production and exchange (n = 18; Table 2). This was highlighted by one participant who said that: ‘Transparency is hugely important, and it will become more and more so. In this time of populist politicians and fake news, if you cannot demonstrate transparency throughout all parts of the process you will be considered untrustworthy...and it’s not just about perceived transparency, but actual transparency’ (ID 2).

Participants noted that within ICES there has been a significant effort invested towards increasing transparency across all functions and operations. In particular, participants (n = 15) highlighted the importance of the ICES Transparent Assessment Framework (TAF), which is an online open resource of ICES stock assessments for each assessment year, where all data is fully traceable and versioned. Participants also emphasised that transparency can be increased by inviting external stakeholders into the process (n = 14), having a clearly documented process for developing and exchanging knowledge (n = 12), having a code of conduct for all process participants (n = 7) and explicitly acknowledging any limitations of the knowledge generation process (n = 6). However, it should be noted that

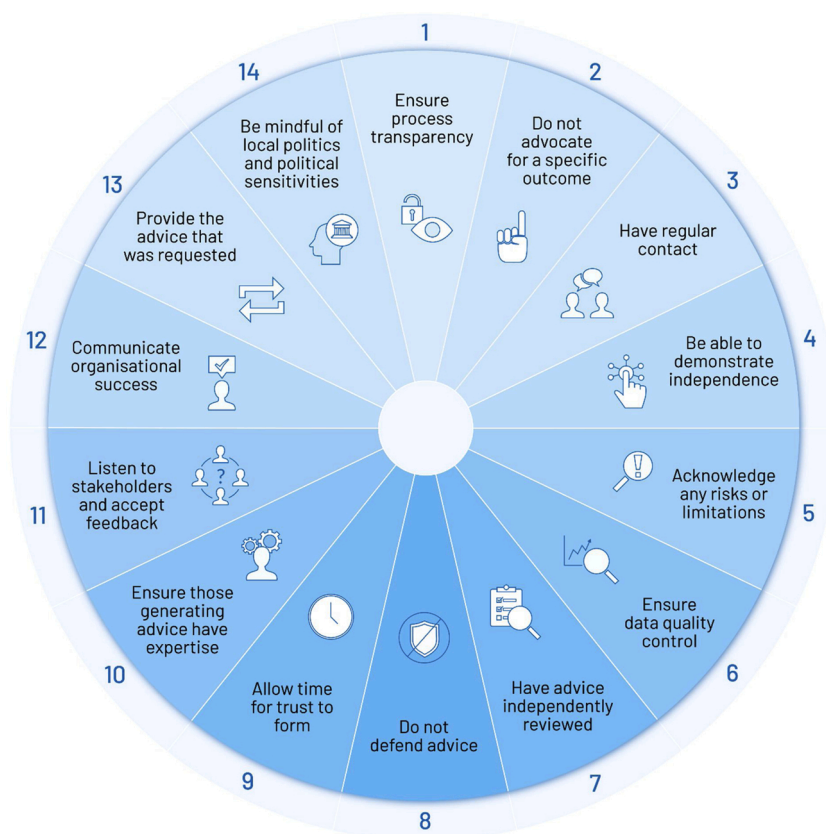


Fig. 1. Fourteen strategies identified through a case study of ICES for building trust at the interface of environmental science and policy.

while transparency was considered critical to trust building, and participants outlined ways to improve transparency, efforts must move beyond simply making the process more accessible to stakeholders, to actually helping them understand the content of the process: ‘...so yeah, transparency in a general sense is good. But there is a misconception that if you open the door to everyone and any time you have transparency. That’s not the case, the people participating also need to understand the context and what is happening’ (ID 10).

Not advocating for a specific outcome was the next most frequently identified consideration when trying to build trust at the interface of environmental science and policy ($n = 14$; Table 2), but rather, remaining impartial. This was highlighted by one participant who said: ‘Our job is not to say what they should do, but to present them with an unbiased summary from the evidence base, explain the potential consequences of any actions they may take based on that evidence, and then let them deal with it as they want’ (ID 2). Participants also highlighted how remaining impartial and not advocating for an outcome is critical for maintaining credibility and legitimacy of the process by all stakeholders, irrespective of their specific values and worldviews. For example, ‘As soon as you start to advocate, then you find yourself being more supported by some stakeholders, but criticised by others for supporting vested interests....this can impact our overall integrity’ (ID 7).

Other commonly discussed strategies for building trust included having regular contact with stakeholders ($n = 7$), being able to demonstrate independence ($n = 9$), and acknowledging any risks associated with the knowledge production process ($n = 8$; Table 2). In relation to regular contact, participants emphasised that trust is most effectively built through regular face-to-face engagement, preferably in both formal and informal settings. The ability to demonstrate independence related to the need to work with numerous stakeholders (e.g. science, policy, industry, non-governmental organisations, etc.), at multiple levels (national, regional, international) and to be able to demonstrate that actions are not driven by the values of goals of any particular stakeholder group. One

mechanism that was identified to support this was having the advice/knowledge independently peer-reviewed before providing it to end-users. Acknowledging risks is closely related to transparency (discussed above), and relates to clearly articulating any limitations associated with the knowledge generation process, such as a lack of data.

Building on the acknowledgment of risks associated with data collection and knowledge generation, participants in our study also identified the importance of having mechanisms in place to ensure the data quality within ICES processes ($n = 7$; Table 2). This was highlighted by one participant who stated that: ‘I think data control is the most critical aspect [to building trust]....data availability, data uncertainty, data quality....this is the underlying key to what we do’ (ID 15). Participants also noted that trust can be built by listening, accepting and actioning feedback from stakeholders (e.g. on relation to approaches to improving knowledge exchange processes), by ensuring that those responsible for generating the advice/knowledge have established and recognised expertise in the topic area, and by being mindful of local politics and political sensitivities when undertaking engagement activities and framing the advice for what could be pragmatically achievable within those contexts (Table 2). Four participants also noted that building trust (and implementing the strategies outlined in this section) requires considerable time: ‘We are dealing with political issues....the trust you need cannot be made overnight...it is earned over decades’ (ID 4).

Although mentioned less frequently, participants identified eight additional strategies as also important for ICES in terms of building trust: (i) Independent review of advice, (ii) Do not defend advice, explain the advice, (iii) Give it time, (iv) Ensure those generating advice have correct expertise, (v) Listen and accept feedback, (vi) Communicate success, (vii) Deliver what was requested/required, and (viii) Be mindful of politics and political sensitivities (see Table 2).

Table 2

The fourteen strategies to trust building based on the experiences of ICES, and specific actions for achieving each one, as derived from qualitative analysis of the interview data (n = 20).

Strategy	Specific actions
1 Ensure process transparency	- Have a clearly documented process for how advice will be generated; - Ensure data is open, traceable and accessible; - Invite external stakeholders into process; - Establish a Code of Conduct for all participants.
2 Do not advocate for a specific outcome	- Present unbiased summary and potential consequences; - Refrain from applying individual values and worldviews; - Ensure impartiality at all times.
3 Have regular contact	- Where possible, utilise face-to-face engagement; - Create opportunities for both formal and informal interactions.
4 Be able to demonstrate independence	- Engage equally with all relevant stakeholders; - Do not favour the perspectives of a specific group of stakeholders; - Declare any potential or perceived conflict of interests; - Where possible, have scientific advice independently reviewed.
5 Acknowledge any risks or limitations	- Clearly articulate any limitations associated with processes of knowledge generation or exchange; - Be explicit about potential uncertainty within the data and associated risks and limitations.
6 Ensure data quality control	- Ensure mechanisms are in place to ensure data quality; - Be explicit about which data was used; - Ensure data is open access, traceable, version controlled and accessible.
7 Have advice independently reviewed	- Establish a formal process for having advice independently peer-reviewed; - Ensure peer reviewers have suitable and recognised expertise for the content; - Establish mechanisms to ensure peer-review comments are adequately accounted for and addressed.
8 Do not defend advice	- Remain impartial at all times; - Explain the advice and all risks/limitations, but do not defend the advice; - Listen and accept feedback about the process of knowledge generation and exchange from stakeholders.
9 Allow time for trust to form	- Trusted relationships are founded upon mutual respect; - Ensure relationships are not just transactional; - Allow sufficient time to build trusted relationships; - Ensure regular contact over time to maintain trusted relationships once established; - Do not take trusted relationships for granted.
10 Ensure those generating advice have expertise	- Build broad teams with relevant disciplinary expertise; - Ensure diversity within teams in terms of career stage, gender and nationality; - Be open to new collaborations and the inclusion of different knowledge systems, where appropriate.
11 Listen to stakeholders and accept feedback	Engage with stakeholders early in the process;

Table 2 (continued)

Strategy	Specific actions
	- If possible, co-create questions together; - Ensure mechanisms are in place to mitigate potential power imbalances between those generating and 'using' the advice; - Include specific mechanisms to receive feedback from stakeholders throughout process.
12 Communicate organisational success	- Document 'successes' as they occur; - Share successes among stakeholders; - Whilst sharing success remain humble, and recognise that there is always room for improvement.
13 Provide the advice that was requested	- Ensure that a clear question is in place to guide the process of generation of knowledge and advice (co-develop this question where possible); - Set joint expectations about what can feasibly be delivered against these questions based on available knowledge and resources; - Ensure regular contact throughout the process regarding progress, limitations and potential risks; - Ensure that the knowledge generated directly answers the question that was originally asked.
14 Be mindful of local politics and political sensitivities	- Be informed of regional and relevant politics; - Be respectful of how local politics and political sensitivities might hinder the ability of stakeholders to use the knowledge that has been generated.

3.3. Trust repair

As detailed in section 3.1, while participants of our study considered trust as critical for effective knowledge exchange at the interface of environmental science and policy, results also highlighted the dynamic and fragile nature of trust. Further, participants noted that whilst trust requires protracted timeframes to form, it can be lost far more quickly, for example, through both intentional and unintentional actions (e.g. the accidental provision of wrong information). Reflecting on their involvement in ICES, and particularly situations where trust with decision-makers had been compromised in the past, participants identified five key steps to trust repair (Fig. 2), as detailed here.

The first step of trust repair identified through data analysis is the need to be honest when a mistake occurs, and advising your stakeholders as quickly as possible (n = 12, Fig. 2). This is explained by one participant who stated: '*When you make a mistake you need to be honest about it...you have to be open to [your stakeholders]...and let them know as soon as possible*' (ID 8). In communicating that a mistake has occurred to stakeholders, participants of our study suggested that it is important to not be defensive and to not seek to blame any individual or organisations. Rather, this step is simply focused on being transparent that an error had occurred.

The second and third steps of trust repair identified through this study are the need to identify why the mistake occurred, and then implementing measures to make sure that it does not happen again, respectively (Fig. 2). As highlighted by one participant: '*We need to be absolutely sure what the mistake was, why it happened, and that it's not going to happen again*' (ID 12). A critical consideration across both of these steps is the need for transparency (in terms of what has happened, and the specific measures implemented to prevent it from happening again), and speed (in terms of identifying the cause of the problem and rectifying it). Additionally, in the third step, results emphasize the importance of communicating process changes to stakeholders, and that this is best done via a face to face format.

Once measures have been put in place to prevent the mistake happening again, the fourth step to repair trust is to correct and reissue

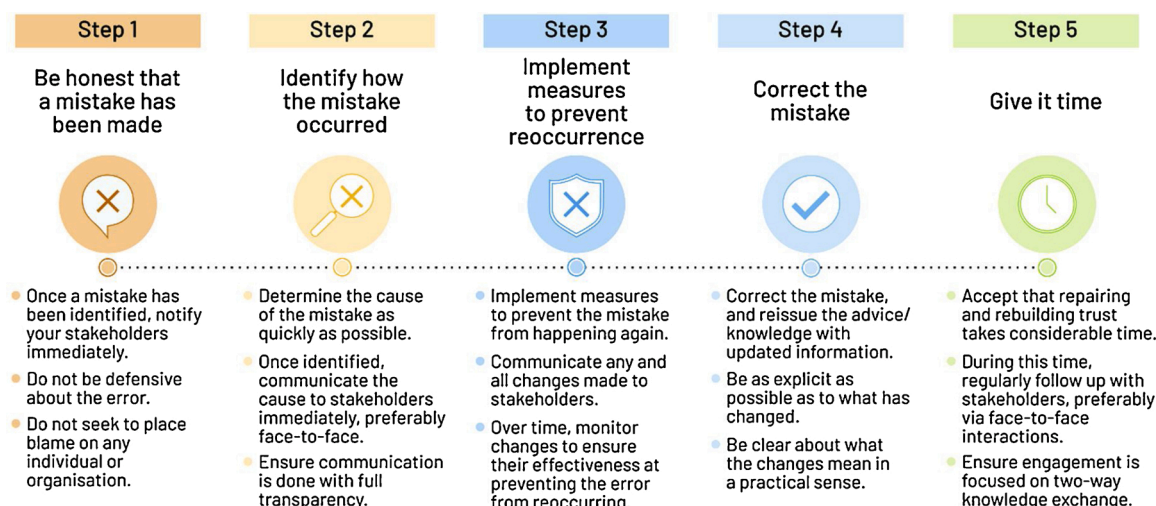


Fig. 2. The five stages of trust repair, and accompanying actions, identified through this study of ICES.

the knowledge/advice that was provided ($n = 6$; Fig. 2). In correcting the advice, it is critical to acknowledge previous errors and to be as explicit as possible as to the changes made, and what they mean in a practical sense. As highlighted by this participant: ‘*When the advice is corrected, it’s not only a change in a figure or a number [in this case the participant was referring to suggested changes to harvest rates and the financial implications]...these changes have a lot of importance to industry and other stakeholders, so we need to be really clear on what has changed – we need to be as transparent as possible on this*’ (ID 8).

Finally, once the previous steps have been implemented, participants in our study identified the importance of time (step 5) for repairing and rebuilding trust ($n = 4$; Fig. 2). In particular, participants emphasized that once trust had been compromised between individuals it would require long timeframes to return to previous levels, and that during this time, it is important that knowledge providers continually and regularly follow up with their stakeholders. Although not a strategy in itself, a recurrent theme in relation to trust repair was the importance of face-to-face engagement. It was acknowledged that regular follow-up with stakeholders would be best achieved via face to face interactions focused on two-way knowledge exchange, where possible.

4. Discussion

While trust is recognised as a critical precondition for successful knowledge exchange (e.g. Addison et al., 2013; Clarke et al., 2013; Reed et al., 2014; Cvitanovic and Hobday, 2018), there is a paucity of empirically derived guidance on how trust can be built and managed at the interface of environmental science and policy. This study sought to begin to fill this gap, via an in-depth qualitative analysis of ICES. Reflecting the results of previous studies, we find that trust is an essential component of efficient and effective knowledge exchange within ICES, although highly dynamic and fragile over time. We also identify specific approaches that ICES has used for building trust, and actions for repairing trust when it is compromised. Here, we discuss these results, with a particular focus on advancing the practical knowledge of trust at the interface of environmental science and policy, for improved knowledge exchange and the attainment of evidence-informed decision-making processes.

4.1. Understanding trust

As described in the introduction, trust is defined as a psychological state in which an entity (i.e. a trustor) accepts some level of vulnerability based on a positive expectation of another entity (i.e. a trustee) (Rousseau et al., 1998). Within our case study and the context of ICES, participants identified three key levels of trust at the interface of environmental science and

policy; (i) trust between individuals, (ii) trust in the organisation (i.e. the ICES ‘brand’), and (iii) trust in the process. Of these, trust in the organisation was discussed most frequently, followed by trust in the process.

Across all three dimensions of trust identified, perceived impartiality was considered critical. For example, our results suggest that trust in the organisation (i.e. ICES) and the processes for producing and translating the evidence is determined by the extent to which the organisation is considered independent (i.e. not representing the interest of specific stakeholders; Dankel et al., 2016), and their processes are transparent, open and accessible to stakeholders. At the individual level, participants highlighted how trust can only be built when personal biases are not reflected within the processes of knowledge generation or exchange. Participants in our study discussed how impartiality across all three of these levels can help confer legitimacy and credibility in the scientific advice being generated, which are recognised as key traits that underpin the uptake and integration of science into decision-making (Cash et al., 2003). Thus, these findings highlight the importance of considering and accounting for the multi-dimensionality of trust when seeking to attain knowledge exchange at the interface of environmental science and policy (Stern and Coleman, 2015).

Our results also highlight the highly dynamic and fragile nature of trust at the interface of environmental science and policy. These findings based on the experiences of ICES members emphasise that maintaining a trusted relationship requires ongoing commitment and effort by all parties, and that a trusted relationship should not be taken for granted. They also reflect discussion by Lacey et al. (2017), who consider the dynamic nature of trust by describing four different scenarios over which trust can manifest. These include what they term the ‘ideal’ trusting relationship, which is one that grows rapidly initially, and then plateaus over time, because the trust between the trustor and trustee is unwavering (Lacey et al. 2017). At the other end of the spectrum is the ‘crash and burn’ scenario, which is one in which trust builds initially between the trustor and trustee, but then ‘an event’ causes a complete loss of trust that can never be recovered. Such an event, for example, may include intentionally providing false or misleading information for personal gain (e.g. to acquire increased research funding) which causes damage the credibility of the trustor.

In combination, these results which are based on experiences of ICES members, emphasise that trust, while simple in concept, is complex by nature. Thus, while many studies suggest the need for ‘trust’ at the interface of environmental science and policy, strategies for building and managing trust over time are needed. In the next section, we draw on the experiences of the ICES community to begin to identify possible strategies to help guide trust building initiatives.

4.2. Building and managing trust

In total, our analysis identified 14 strategies that have assisted in trust building within the ICES context. Of these, the most frequently discussed was the need transparency across the entire process of knowledge production and exchange. The importance of transparency reflects discussion by Lacey et al. (2015), who consider the theoretical importance of transparency at the interface of adaptation science and policy, along with the ethical considerations of knowledge exchange. Our results extend that discussion by also identifying specific strategies for achieving this level of transparency.

The most commonly discussed strategy for achieving transparency, as identified through this study, is inviting external stakeholders to participate within the processes of knowledge production and exchange. This is in line with previous research into ICES and fisheries science and management (Wilson, 2009; Glenn et al., 2012). The notion of inviting external stakeholders into processes of knowledge generation and exchange is not new, but aligned with the long-standing ideas associated with participatory research (Lewin, 1946; Tax, 1958). These ideas arose as a response to positivistic notions in the sciences that excluded many voices and sources of knowledge, particularly local, indigenous, gender, and ethnic minority knowledge (Brown and Tandon, 1983; Park et al., 1993; Greenwood and Levin, 2007). The core intent of participatory research approaches is to make research both more democratic, through stakeholder involvement, and impactful, by tying it to actions that benefit those stakeholders (Reason and Bradbury, 2008; Wallerstein and Duran, 2008). Recent work, however, has highlighted how simply ‘opening up a process’ to external stakeholders alone is insufficient for meeting such aspirations, which is also reflected by the results of this study. For example, even when external stakeholders are invited into a process, power imbalances can remain (Few et al., 2007), and in such situations, the more ‘powerful’ individuals or groups can consciously or subconsciously steer participation towards support for a predetermined outcome (Few, 2001, 2003). Thus, while it is beyond the scope of this study to provide guidance on how to identify and manage such power imbalances, we suggest that they must be accounted for to achieve actual transparency, and thus build trust.

Not advocating for a specific outcome was also identified by members of the ICES community as an important consideration when trying to build trust at the interface of environmental science and policy. Indeed, the question of whether or not scientists (or scientific agencies, communities or professional societies) should advocate has been a longstanding and ongoing topic of debate within the academic community (e.g. Nelkin, 1977; Pielke, 2002, 2007; Smith et al., 2008; Kotcher et al., 2017). Much of that debate has centred on the proposition that advocating (or generally participating in advocacy related activities) can compromise the credibility of scientists and their organisations (Nelson and Vucetich, 2009), and thus science and advocacy are inherently discordant (Ruggiero, 2010; Wilson, 2009). In contrast, others have argued that it is impossible to divorce communication and engagement activities from advocacy (e.g. Schmidt, 2015), and that rather than attempting to achieve the impossible (achieving true objectivity), scientists should be more explicit about their personal values and worldviews. A common element across both camps, however, is that advocacy can be defined and viewed in a multiplicity of ways and is highly context specific (e.g. dependent on the goals of individual scientists or scientific institutions). Even so, our results suggest that advocating could have adverse impacts when seeking to build trust for improved knowledge exchange and evidence-informed decision-making processes.

Within our interviews participants did anecdotally note that tensions can arise between opening up a process to be more inclusive and participatory, whilst ensuring that individuals do not advocate for a specific outcome. Whilst it was outside of the scope of this study to explore this tension in depth, this is a key consideration within ICES who use a number of mechanisms to try mitigate this tension. For example, ICES has been influenced by, and uses, all three of the mechanisms

suggested by Rice (2011); (i) peer review of advice by diverse experts, (ii) integrating advice on ecological, economic, and social information and outcomes, and (iii) focusing advisory approaches on risks, costs, and trade-offs of different types of management error. Further, all scientific experts that are engaged in ICES are required to declare any actual, potential or perceived conflicts of interest and state their commitment to abide by the principle of independence in the ICES Code of Conduct. The expert groups are then required to collectively discuss the declaration and ‘police’ adherence to the Codes of Conduct by all individuals within the group. However, given the tension described above, particularly in relation to the perceived independence, this is an ongoing area of consideration and development for ICES, and one that we feel warrants additional direct consideration in the literature in the future.

Another recurrent theme throughout the strategies for building trust within the context of ICES is the importance of face-to-face engagement. However, while face-to-face interactions may represent the most effective pathway towards building trust (and particularly at the individual level), they are also inherently time consuming, costly, and at times, unachievable (Cvitanovic et al., 2019). For example, if knowledge ‘providers’ and their stakeholders are geographically isolated it makes face-to-face interactions particularly challenging. Face to face engagement can also be interpreted as undermining transparency unless clear and open records are maintained for others to access. Indeed, the current study, including all data collection, was undertaken during the global Covid-19 pandemic, which severely impacted the ability for face-to-face meetings globally and for a protracted time (still ongoing at the time of writing this paper). Thus, in considering our list of strategies for building trust, it is important to consider how they may work in virtual settings.

Building trust within virtual teams has been a topic of continued investigation within the business and performance management sectors (e.g. Kanawattanachai and Yoo, 2002; Coppola et al., 2004; Greenberg et al., 2007). This body of literature has identified that key barriers to creating trust in virtual settings include the lack of opportunity for individuals to have the level and depth of dialogue that is necessary to create a shared future, a shortage of time for informal and generative conversations that promote commitment to one another, and the lack of opportunity for personal conversations such as those that encourage self-disclosure and sharing of feelings (Holton, 2001). Indeed, a recurrent theme throughout this literature is the need for informal so-called ‘water-cooler chats’ for building trust (Zakaria, 2008), and the need to create ‘safe spaces’ and dedicated time for such conversations within virtual settings.

In addition to identifying strategies and considerations when working to build trust, our results also provide insights into how trust can be repaired in cases where it has been compromised. Once again, drawing on the business and organizational management literature (reviewed by Lewicki and Brinsfield, 2017), evidence suggests that broken trust can be repaired (e.g. Mishra, 1996; Bottom et al., 2002). Our findings from the analysis of ICES also reflect this, suggesting that damaged trust at the interface of environmental science and policy can also be rebuilt. However, the success of this trust repair is dependent on the response of the trustee following an indiscretion (Korsgaard et al., 2002; Schweitzer et al., 2006), and in most cases, trust repair actions are implemented too slowly and with inappropriate equivocation in an attempt to conceal the truth. To that end, Gillespie and Dietz (2009) outline a model of trust repair consisting of four key steps: (i) immediately acknowledge the mistake to stakeholders, (ii) systematically diagnose the cause of the error, (iii) implement interventions to correct the error, and (iv) monitor the effectiveness of the interventions. The process of trust repaired identified through our case study of ICES largely aligns with this broad structure, while adding a fifth step that explicitly recognises the time and effort that will be required to build trust, and specific actions that can be taken during this time to facilitate the re-building of trust.

4.3. Limitations and future research

In this study we have begun to develop a deeper and empirically grounded understanding of trust at the interface of environmental science and policy. There are, however, important limitations to our study which should be considered. Reflecting on these limitations also helps to identify a roadmap for future research into trust at the interface of environmental science and policy.

First, the strategies for building and managing trust as derived through this study are based on the experiences and perceptions of the ICES community. As such, while we were able to elucidate a range of strategies and considerations, we are unable to assign relative importance to each of these in practice, or understand the extent to which each confers trust among different ‘end-users’. Thus, future research should also seek to elucidate the perspectives of different end-users so as to understand their perceptions about which strategies are most effective in terms of building and managing trust.

Next, the strategies and considerations presented in this study have been derived from a single case study. While ICES represents a strong case study to explore notions of trust, we acknowledge that the interface between environmental science and policy is highly context specific across cultures, space and time. Therefore, future research is also needed to understand the applicability of these findings in other contexts, and in doing so, help to develop a more representative and comprehensive set of strategies for building and managing trust at the interface of environmental science and policy.

In addition, this study supports previous studies to highlight the dynamic and fragile nature of trust over time. In doing so, it also highlights the importance of being able to measure and monitor trust over time to ensure sustained and effective knowledge exchange. While it was outside of the scope of this study to begin to understand methods and indicators that can be used to measure and monitor trust over time, this should also be pursued through future research effort.

Finally, throughout this study we have focused on the benefits of trust, particularly as they relate to knowledge exchange at the interface of environmental science and policy. We acknowledge, however, that previous work has highlighted the risks associated with trust, and particularly, too much trust. For example, Lacey et al. (2018) outline a situation that they refer to as ‘blind faith’, whereby a decision-maker may trust an individual scientist so much that they do not look for signs of misconduct, such as the misrepresentation of findings. This can not only have detrimental impacts to the credibility and legitimacy of both scientists and decision-making agencies (Cvitanovic et al., 2019), it could also have adverse flow-on effects to the sustainable management of environmental resources. Thus, while outside of the scope of this study, we believe that additional research is needed to more comprehensively understand the risks associated with trust at the interface of environmental science and policy, with a specific focus on identifying mechanisms to mitigate those risks across contexts.

5. Conclusion

Through this analysis of ICES we have highlighted the importance and fragility of trust at the interface of environmental science and policy, and identified 14 strategies with key actions to establish and nurture trust between trustor and trustee. Despite the best intentions, however, there are circumstances where trust can be lost or eroded, therefore we also provide key steps for trust repair. In presenting these insights, this research seeks to provide practical and implementable strategies that can be used by environmental scientists and/or research institutions to help them foster trust with decision-makers, enable improved knowledge exchange and build capacity for evidence-informed decision-making processes.

Author statement

CC, MD-C and IvP conceived and designed the study. Data collection was undertaken by CC, MM and RS. Data analysis was led by CC,

with input from MM, RS and DBK. Data interpretation was undertaken by all authors. All authors contributed to the writing and revision of the manuscript.

Declaration of Competing Interest

The authors report no declarations of interest.

Acknowledgements

We thank the ICES Advisory Committee (ACOM) leadership for their support in developing this study, and all participants who took part in this study. We also thank S. A. McCormack (Visual Knowledge) for the development of Figs. 1 and 2. Finally, we thank the three reviewers who provided helpful and constructive feedback on our paper.

Appendix A

Interview Guide

Introductory questions

- 1 What is your role in [organisation name]?
a Prompt – [If not ICES] What is your role/relationship to ICES?
- 2 How long have you worked in this role?

Understanding trust

- 3 To what extent do you think trust in ICES as a provider of scientific knowledge from the requesters of ICES advice is important?
- 4 To what extent do you think ICES has been successful at establishing trust with advice requesters?
- 5 How do ICES establish, maintain and grow trust with the requesters of advice?
- 6 Can you think of a time where trust may have been compromised or lost? How (if at all) was it recovered?
- 7 What do you consider as the critical things for building and maintaining trust with the requesters of advice?
- 8 How important is transparency when providing advice to requesters?
a Prompt: What measures are in place to ensure transparency?
b Prompt: How do ICES ensure their credibility and legitimacy?

Conclusion

- 9 Now that we have completed the formal component of the interview, are there other important issues that were not covered by our questions, or other relevant insights that you would like to share based on your experience in ICES?
- 10 And finally, considering the topic of the study, and the questions I’ve just asked you, who are three other people that you think I should interview in this research?

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